EXPERIMENT NO 04

18-10-2026

NAME: YUGITHA B

REG NO: 192424102

SLOT: A

SUBJECT CODE: CSA4701

SUBJECT: DEEP LEARNING

SEMESTER: MS2-SEPTEMBER-2026

Entropy, Cross-Entropy and KL Divergence

SCENARIO

A decision-support system classifies transactions into three risk categories. Calculate the entropy of the predicted class distribution to measure its uncertainty.

AIM

To calculate the entropy of a predicted transaction risk-class distribution using Python and NumPy to measure uncertainty in a decision-support system.

ALGORITHM

1. Import the required libraries.
2. Define the predicted probabilities for three risk categories.
3. Calculate entropy using the probability distribution.
4. Display the probabilities and entropy value.
5. Visualize the class distribution using a bar chart.

SOURCECODE

```
import numpy as np
import matplotlib.pyplot as plt

# Predicted probabilities for three risk categories
categories = ['Low Risk', 'Medium Risk', 'High Risk']
probabilities = np.array([0.5, 0.3, 0.2])

# Calculate entropy
entropy = -np.sum(probabilities * np.log2(probabilities))
```

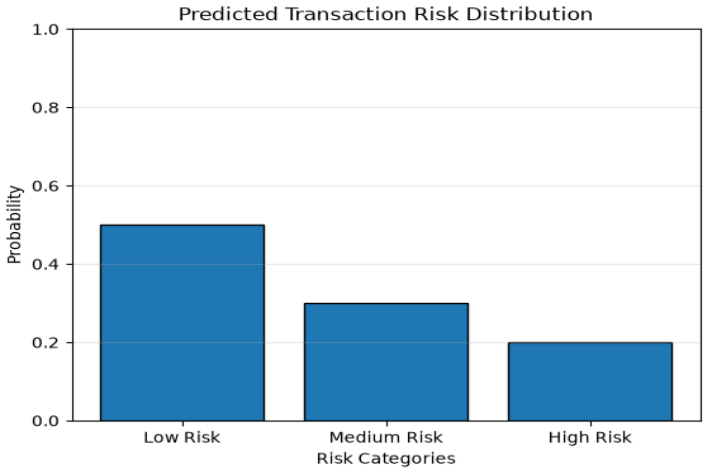
```
# Display results
print("Transaction Risk Distribution")
for category, probability in zip(categories, probabilities):
    print(f"{category}: {probability:.2f}")
print("\nEntropy:", round(entropy, 4), "bits")

# Visualize probability distribution
plt.bar(categories, probabilities, edgecolor='black')
plt.title('Predicted Transaction Risk Distribution')
plt.xlabel('Risk Categories')
plt.ylabel('Probability')
plt.ylim(0, 1)
plt.grid(axis='y', alpha=0.3)
plt.show()
```

OUTPUT

```
Transaction Risk Distribution
Low Risk: 0.50
Medium Risk: 0.30
High Risk: 0.20

Entropy: 1.4855 bits
```



RESULT

Thus, entropy was calculated successfully to measure uncertainty in the predicted transaction risk distribution. The summary table and graphs illustrate the probability of each risk category and its contribution to the overall uncertainty.